

Harnessing slaughterhouse by-products: From wastes to high-added value natural food preservative



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ABSTRACT

Blood, from slaughterhouses, is an inevitable part of meat production, causing environmental problems due to the large volumes recovered and its low valorization. However, the α 137–141 peptide, a natural antimicrobial peptide, can be obtained after hydrolysis of hemoglobin, the main constituent of blood red part. To recover it at a sufficient concentration for antimicrobial applications, a new sustainable technology, called electrodialysis with ultrafiltration membrane (EDUF), was investigated. The α 137–141 concentration was increased about 4-fold at a feed peptide concentration of 8% with an enrichment factor above 24-fold. This feed peptide concentration also needed the lowest relative energy consumption. Moreover, this peptide fraction protected meat against microbial growth, as well as rancidity, during 14 days under refrigeration. This peptide fraction was validated as a natural preservative and substitute for synthetic additives against food spoilage. Finally, producing antimicrobial/antioxidant peptide from wastes by EDUF fits perfectly with the concept of circular economy.

1. Introduction

Blood from slaughterhouses is an inevitable part of the meat production causing environmental problems, due to the very large volumes recovered and its low valorization (Hoyo, Rendueles, & Diaz, 2008). Indeed, in 2016, 30 million litres of bovine and porcine blood were produced in the Quebec province and > 120 million in Canada (Canada, 2016). The colourless part of blood, called plasma, already has applications, mainly by protein recovery for the pet-food and pharmaceutical sectors (Toldrà, Aristoy, Mora, & Reig, 2012), but the cellular red fraction of blood, the cruor, representing about 40% of whole blood, is an underutilized material despite its wealth of bioactive compounds (Lynch, Mullen, O'Neill, & Garcia, 2017). This red part contains mainly hemoglobin (> 90%), a protein widely described as a rich source, after its hydrolysis, of antimicrobial peptides (Nedjar-Arroume et al., 2008). One of these antimicrobial peptides, the α 137–141 peptide (653 Da, pI of 10.5), has been recently reported as a natural promising candidate for the substitution of synthetic additives used to protect meat during its storage and distribution (Przybylski, Firdaous, Châtaigné, Dhulster, & Nedjar, 2016).

This finding is particularly interesting because the waste from

slaughterhouse by-products could be a potential natural preservative. Food waste bioconversion could reduce the environmental impact of industrial activities by minimizing synthetic additive production. Moreover, this way of waste transformation improves the development of food production sustainability by harnessing by-products (Goot et al., 2016).

However, the main challenges in the production of such a bioactive peptide are its isolation from the complex feed hydrolysate and its recovery concentration in order to be effectively bioactive (Agyei & Danquah, 2011). Some approaches have proposed the use of pressure-driven processes or solvents to extract the peptide fraction from the hydrolysate containing more than one hundred peptides (Agyei, Ongkudon, Wei, Chan, & Danquah, 2016). Nevertheless, pressure-driven processes are not able to separate compounds with similar molecular weights because they have a weak selectivity. Moreover, many methodologies are discouraged for food application when solvents or toxic chemicals are used, e.g. chromatographic processes or liquid-liquid separations. In addition, this kind of process is an expensive approach for production of peptide fractions, generally with no-circular economy for the recycling of solvents. Recently, some studies have shown successful results for peptide separation from by-product

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hydrolysates from different sources, such as snow crab (Koumfieg Noudou et al., 2016) or rapeseed (He et al., 2016), with a new green technology generically named electrodialysis with filtration membrane (EDFM) or more specifically electrodialysis with ultrafiltration membrane (EDUF). This technology separates peptides according to their charge (by application of an electrical field) and their molecular weight (molecular weight cut-off of the filtration membrane) (Bazinet, Amiot, Poulin, Tremblay, & Labbé, 2005).

Our proposed approach is to use this hybrid technology to recover, from cruor hydrolysate, a peptide fraction containing the α 137–141 antimicrobial peptide in sufficient concentration to be applied as a meat preservative. However, for its potential application, it is important to separate this peptide at a high purity and concentration. In this context we propose to investigate the impact of the feed peptide concentration. To the best of our knowledge, no study has been reported about the impact of feed concentration on the recovery of a specific peptide and peptide migration behaviour during EDUF separation. Following the α 137–141 separation, the recovered EDUF peptide fraction was tested on ground beef to evaluate its potential as an antimicrobial and anti-oxidant agent for food protection.

2. Materials and methods

2.1. Chemicals

All chemicals and solvents were of analytical grade and from commercial suppliers: Sigma-Aldrich (Saint-Quentin Fallavier, France) or Flandres Chimie (Villeneuve d'Ascq, France). Bovine cruor was provided by Vapran SA (Pleinet, France). Ultrapure water was prepared using a Milli-Q system.

2.2. Cruor hydrolysis

A stock solution was prepared by adding 15 g of cruor to 100 ml of ultrapure water. The real bovine hemoglobin concentration (C_{BH}) was dosed by the Drabkin's method (Crosby, Munn, & Furth, 1954), after 30 min of centrifugation at 4000 rpm. After that, the stock solution was diluted to obtain a solution with a given C_{BH} (1, 2, 5 and 8% w/v). The concentration of 8% (w/v) was the maximum to keep the homogeneity and the cruor solubility in the hydrolysate without salt addition.

Hydrolysis was carried out at pH 3.5 at a constant temperature (23 °C) with an enzyme/substrate ratio of 1/11 (mole/mol) (Dubois, Nedjar-Arroume, & Guillochon, 2005). Bovine cruor was digested by pepsin from porcine gastric mucosa (EC 3.4.23.1 manufactured by Sigma-Aldrich, 3200–4500 units.mg⁻¹ protein) and the hydrolysis was stopped by NaOH (5 M) up to a final pH of 9 after 30 min, corresponding to a hydrolysis degree of 5%, assessed by the *ortho*-phthalaldehyde method (Church, Swaisgood, Porter, & Catignani, 1983). Indeed, in a preliminary study, this hydrolysis degree showed a higher migration of α 137–141 peptide in terms of purity and concentration in the recovery compartment.

After hydrolysis, the hydrolysate was decolorized in order to eliminate the remaining haem since it was shown in a previous study that the presence of haem has a negative impact on the peptide migration during EDUF (Vanhoute et al., 2010). The haem was precipitated and peptides were maintained in the solution. Both phases were separated by centrifugation (30 min, 4000 rpm) and the supernatant was kept as feed hydrolysate.

2.3. Electroseparation

2.3.1. Electrodialysis with ultrafiltration membrane equipment

The electrodialysis cell was a MicroFlow type cell (Electrocell AB, Täby, Sweden) with an effective area of 10 cm². The cell configuration is displayed in Fig. 1. The cationic configuration was used to allow the specific recovery of positively charged compounds such as the

α 137–141 peptide and to remove most of the other peptides, mainly negatively charged peptides, which could interfere with the desired activity.

Four closed compartments were delimited by the different membranes. In this configuration, the anionic and cationic membranes separated the Na₂SO₄ (20 g.l⁻¹) electrode rinsing solution. At the beginning of the process, on both sides of the ultrafiltration membrane, the cationic peptide recovery and the feed compartments were circulated and contained, respectively, a KCl solution (2 g.l⁻¹) and a hydrolysate, freshly prepared at a given C_{BH} . Each compartment was connected to a separate external reservoir to allow the recirculation of the solution by using centrifugal pumps (Iwaki, Marcoussis, France). The flow rate of each solution was controlled by its own flowmeter (Kobold Instrumentation, Cergy-Pontoise, France). Finally, the anode/cathode voltage difference was supplied by a variable 0–30 V power source HQ Power PS3003 (Xantrex, Burnaby, Canada).

The ultrafiltration membrane (UF) used was made of polyethersulfone (PES) with a molecular weight cut-off of 10 kDa (HFK-131, Koch, France). The ion-exchange membranes were Neosepta cationic (CMX-SB) and anionic (AMX-SB) food grade membranes manufactured by Astom (Tokyo, Japan) and provided by Eurodia (Pertuis, France).

2.3.2. Experimental procedure

All the compartments contained 500 ml of solution (Na₂SO₄, KCl or hydrolysate). The voltage was maintained constant throughout the process at 20 V, corresponding to an electric field of 9 V.cm⁻¹. The circulation flows were of 18 l.h⁻¹ for the electrode solution (Na₂SO₄) and of 12 l.h⁻¹ for the recovery compartment (KCl) and the feed compartment (hydrolysate).

During experiments, each electroseparation at different feed peptide concentrations was carried out in triplicate. Membranes in the EDUF cell were all changed for a new tested concentration.

During electroseparation, the pH was maintained constant at a given value of 9 by addition of HCl (0.1 M) or NaOH (0.1 M) in the hydrolysate and KCl compartments. Preliminary studies showed that this pH value allowed the best enrichment factor in α 137–141 by minimizing the number of peptides which co-migrated. The EDUF separation was carried out for 2 h and, after each repetition, the EDUF cell was cleaned to remove a potential deposit in the apparatus.

The pH, conductivity and current intensity were recorded throughout the treatments under the different conditions of feed peptide concentration. The relative energy consumption was evaluated for each condition. Furthermore, the final fractions containing the antimicrobial peptides were analyzed in terms of peptide profile and sequence identification.

Among the four recovered peptide fractions, the one containing the highest concentration of α 137–141 was applied to meat, to assess its potential and test its food preservative properties.

2.4. Evaluation of meat preservation

The microbial growths and the lipid oxidation were evaluated during 14 days of meat preservation under refrigeration (4 °C) according to the protocol previously described (Przybylski et al., 2016).

Briefly, the experiment was repeated, at least in triplicate, with different commercial beef patties purchased on different dates to ensure the preservation reproducibility. All experiments were carried out under aseptic conditions. Three conditions of storage were studied: (1) meat without additive being the negative control; (2) meat with the α 137–141 standard peptide being the positive control and (3) meat incorporating the EDUF peptide fraction. The preservative powder was added to the meat by blending. The selected ratio, preservative/meat was 1/20.

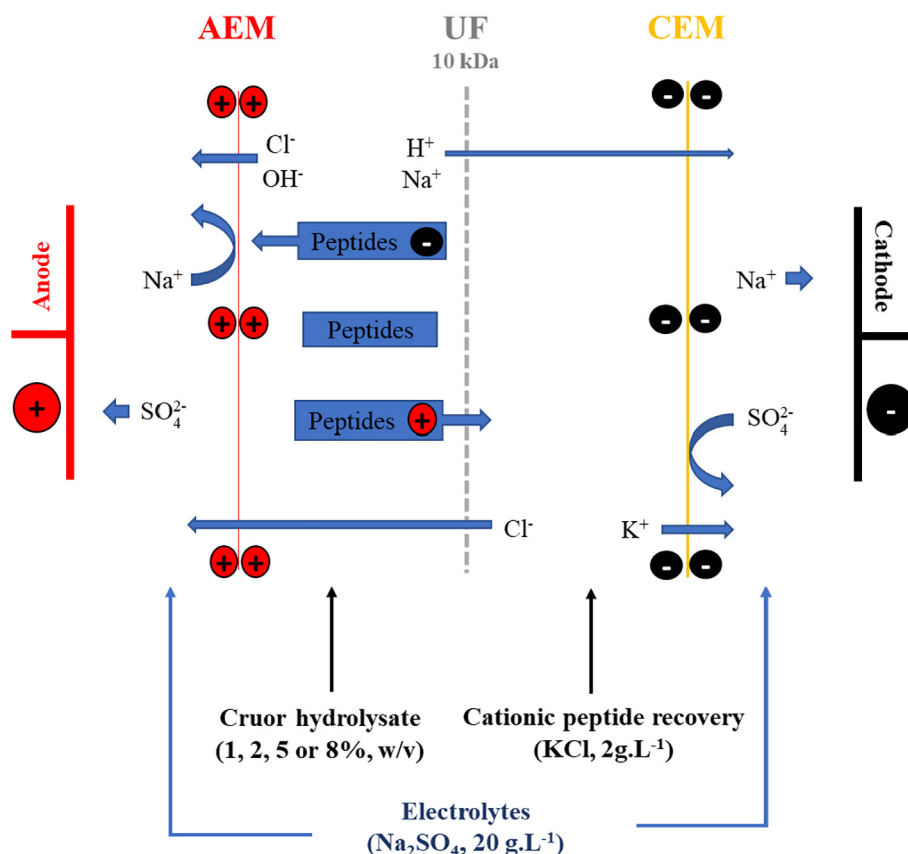


Fig. 1. Schematic representation of electrochemical cell with cationic configuration. CEM: cationic exchange membrane; AEM: anionic exchange membrane; UF: ultrafiltration membrane.

2.5. Analytical methods

2.5.1. Relative energy consumption

The total energy consumption (E_T) during the electroseparation was assessed by the following equation (Koumfieng Noudou et al., 2016):

$$E_T = U \cdot \int_{t=0}^{t=2h} I(t) dt \quad (1)$$

where E_T (Wh) is the energy, I (A) the current intensity and U (V) the voltage.

The relative energy consumption (E_R) was expressed in Wh/g of total peptide recovered and calculated according to the following equation:

$$E_R = \frac{E_T}{\text{Amount of peptides}} \quad (2)$$

where the amount of peptides (g) is the quantity of recovered peptides at the end of the EDUF treatment.

2.5.2. RP-HPLC analyses and quantification method for total peptides and $\alpha 137-141$ peptide

The liquid chromatographic system consisted of a Waters 600E automated gradient controller pump module, a Waters Wisp 717 automatic sampling device and a Waters 996 photodiode array detector (Milford, USA). The applied methodology for analysis and peptide quantification, mainly $\alpha 137-141$, was described in a previous paper (Przybylski et al., 2016).

Additionally, the total peptide concentration of each sample from the BH hydrolysate was evaluated by measuring the total area corresponding to the peptides (A_{tot}) with the Millennium software. This measurement was correlated with the initial C_{BH} measured by using the Drabkin method (Crosby et al., 1954). The measured peptide

concentrations were converted into peptide migration rates ($\varphi_{\text{peptides}}$ expressed in $\text{g}_{\text{peptide}} \cdot \text{h}^{-1} \cdot \text{m}^{-2}$) as follows:

$$\varphi_{\text{peptides}} = \frac{\text{Amount of peptides}}{\text{treatment duration} \times \text{membrane surface area}} \quad (3)$$

where amount of peptides is the recovered peptides at the end of treatment (g), treatment duration being the total time of EDUF operation (h) and the surface membrane area being expressed in m^2 .

The $\alpha 137-141$ purity ($P_{\alpha 137-141}$ expressed in %) was evaluated by:

$$P_{\alpha 137-141} = \frac{A_{\alpha 137-141}}{A_{\text{total}}} \quad (4)$$

where $A_{\alpha 137-141}$ is the area of the $\alpha 137-141$ peak and A_{total} is the total area of all peptide peaks, both measured on the chromatogram.

2.5.3. Mass spectrometry analyses

The LC-MS analyses were performed on a UFLC-XR device (Shimadzu, Japan) coupled to a QTRAP® 5500 MS/MS hybrid system triple quadrupole/linear ion trap mass spectrometer (AB Sciex, Foster City, CA, USA) equipped with a Turbo VTM ion source. Instrument control and data acquisition were performed using the Analyst 1.5.2 software. The RPLC separation was carried out on the same column as used for the RP-HPLC analyses (§ 2.5.2.). The elution was performed with the same gradient, previously described, using formic acid (0.1%) instead of trifluoroacetic acid (0.1%) for the RP-HPLC analyses. MS analysis was carried out in positive ionization mode, using an ion spray voltage of 5500 V. The nebulizer gas (air) and the curtain gas (nitrogen) flows were set at 30 psi. The Turbo VTM ion source was set at 550 °C with the auxiliary gas flow (air) set at 50 psi.

The MS/MS analyses were completed with the BioAnalyst 1.5.1 and Peaks 7 softwares.

2.5.4. Lipid oxidation evaluation

Meat rancidity is mainly due to lipid oxidation during the storage of food. A preservative should be an excellent protector against this phenomenon to avoid organoleptic alteration and to increase the meat shelf-life (Kim, Cho, & Han, 2013; Weiss, Gibis, Schuh, & Salminen, 2010). The lipid oxidation was quantified by measurement of the thiobarbituric acid-reactive substances (TBARS) (Witte, Krause, & Baily, 1970). The protocol is adapted from a previous study (Przybylski et al., 2016). Briefly, 5 g of meat were added to 12.5 ml of trichloroacetic acid solution (20% w/v) and phosphoric acid (2 M). The solution was transferred to 25 ml of ultrapure water and filtered; 5 ml of this filtrate were added to 5 ml of 2-thiobarbituric acid (0.005 M). After 17 h at room temperature, the results were read at an absorbance of 520 nm and expressed as malondialdehyde equivalents per kilogram of meat (mg MDA/kg meat).

2.5.5. Microbiological analyses

Counts were carried out at days 0, 2, 4, 7, 9, 11 and 14 of storage at 4 °C, using the same protocol as previously described (Przybylski et al., 2016). Briefly, 5 g of beef patty were added to 45 ml of sterilized peptone water (0.1% w/v) in a sterile stomacher bag and were homogenized (2 min at 200 rpm). Aliquots in peptone water (0.1 w/v) were prepared at decimal dilutions (10^{-1} to 10^{-5}) and 100 µl of the appropriate dilutions were spread out on different selective media: (1) Plate Count Agar (PCA) for total viable counts; (2) Potato Dextrose Agar (PDA) for yeasts and molds counts; (3) Man, Rogosa and Sharpe (MRS) agar for lactic bacteria; (4) Violet Red Bile Lactose (VRBL) agar for coliform counts. All agar plates were incubated at 37 °C for 24 h. The results were expressed as the survival rate calculated as follows: $\log_{10} \frac{N_t}{N_0}$ where N_t is the count at a given storage day and N_0 is the initial count at the beginning of the experiment.

2.5.6. Statistical analyses

One-way analyses of variance (ANOVA) and Tukey tests, using SigmaPlot software (version 12.0), were carried out to determine which condition was significantly different from the others with a significant p -value > 0.05.

3. Results and discussion

3.1. Electrodialytic parameters' evolution

For the hydrolysate compartment, the initial feed conductivities were not significantly different whatever the feed concentration tested with an averaged value of $14.01 \pm 0.38 \text{ mS.cm}^{-1}$. The four feed conductivities evolved similarly during the EDUF treatment to obtain a final conductivity of about $16.14 \pm 0.51 \text{ mS.cm}^{-1}$ (Fig. 2a).

By contrast, the conductivity of the cationic peptide recovery compartment decreased throughout the EDUF treatment and that whatever the tested concentration (Fig. 2b). For the 1% peptide concentration, the conductivity curve shape of the peptide recovery compartment was similar to that of the 2% peptide concentration with an averaged starting value of 2.78 mS.cm^{-1} and an averaged ending value of 1.40 mS.cm^{-1} . However, the conductivity values of the peptide recovery compartment at 5% and 8% peptide concentrations in the feed were significantly higher at the end of the experiment (about 1.80 mS.cm^{-1}). For all tested concentrations, a linear shape was noted with an average slope of $-0.012 \text{ mS.cm}^{-1}.\text{min}^{-1}$ ($r^2 = 0.88$).

The observed results were in accordance with the used cell configuration. Indeed, the demineralization of cationic peptide recovery compartment was mainly due to the migration of K^+ and Cl^- from the KCl solution to the electrolytes on the cathode side and to the feed, respectively (Firdaus et al., 2009). By contrast, the hydrolysate mineralization was due to the migration of Cl^- from the KCl solution during the EDUF treatment (Doyen, Saucier, Beaulieu, Pouliot, &

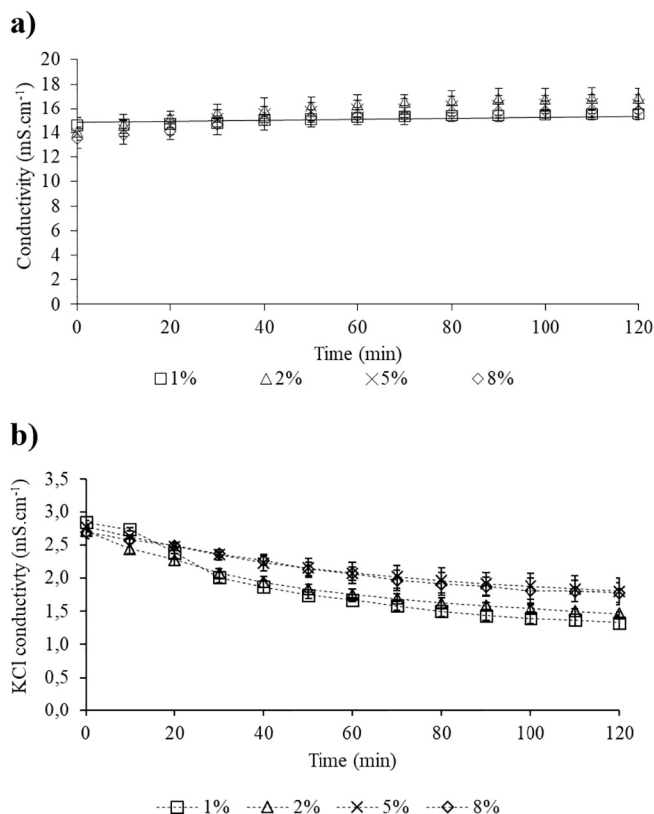


Fig. 2. Evolutions of the hydrolysate conductivities (a) the KCl conductivities (b), the current intensity (c), the total peptide migration (d), the α 137–141 peptide migration (e) and relative energy consumption (f) during the EDUF treatment.

Bazinnet, 2012).

The current intensity, during the electrosorption was not significantly different whatever the feed concentration tested or the experimental duration (Fig. 2c) (Koumfieg Noudou et al., 2016). Globally, the system resistance evolved from 43 ± 1.11 to $222 \pm 5.74 \Omega$. The system resistance increase could be explained by the decrease of the current intensity during the treatment in accordance with Ohm's law.

3.2. Feed concentration effect on the total peptide migration

Fig. 2d shows the peptide migrations during the 2 h of EDUF treatment for the four feed peptide concentrations tested. The obtained peptide concentration in the recovery compartment increased with an increase in feed peptide concentration from 6.46 ± 0.72 , 15.2 ± 0.88 , 41.7 ± 2.17 to $76.4 \pm 3.03 \text{ mg.l}^{-1}$, respectively for 1, 2, 5 and 8%. The highest peptide concentration recovered was obtained with the hydrolysates at 8% (w/v) as feed solution: this corresponded to a cationic migration rate from feed to recovery compartment of $18.7 \pm 0.74 \text{ g.m}^{-2}.\text{h}^{-1}$. By comparison with other studies (Koumfieg Noudou et al., 2016; Roblet, Doyen, Amiot, & Bazinet, 2013; Suwal, Roblet, Amiot, Doyen, Beaulieu, Legault, et al., 2014; Suwal, Roblet, Doyen, Amiot, Beaulieu, Legault, et al., 2014), this rate was one of the highest observed in EDUF for peptide selective separation from a complex hydrolysate and it was comparable with the cationic migration rate at 4% feed peptide concentration obtained for a snowcrab by-product hydrolysate reported in a previous study (Koumfieg Noudou et al., 2016). Nevertheless, the initial feed concentration was 8 times (8% w/v) higher than that of the hydrolysate at 1% (w/v) but the total peptide recovery was increased about 12-fold ($76.4 \pm 3.03 \text{ mg.l}^{-1}$). To explain this fact, the chromatographic profiles of the recovery compartment after 2 h of EDUF treatment are displayed in Fig. 3. The first

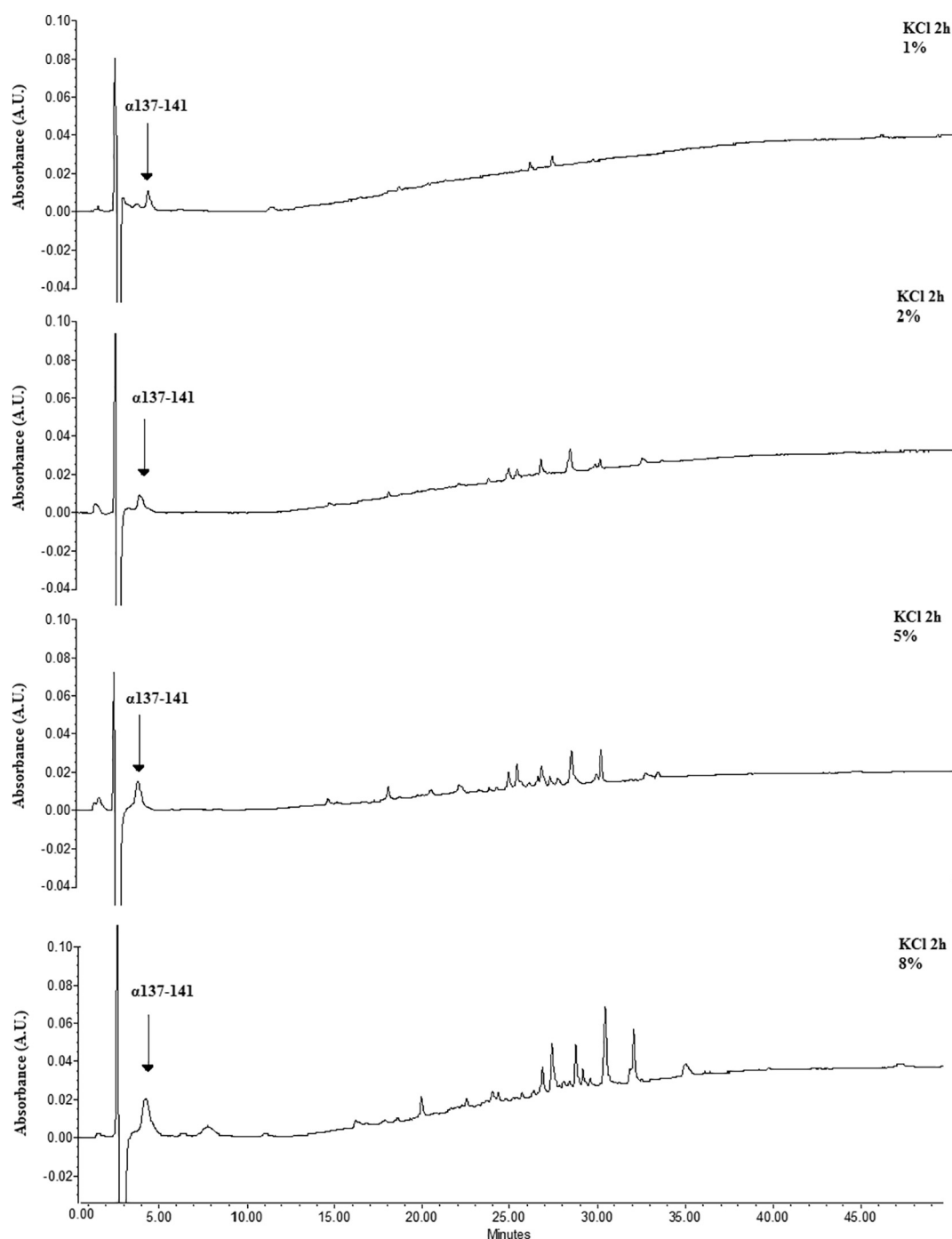


Fig. 3. Chromatographic profiles of total peptide recovery after 2 h of EDUF treatment using different feed peptide concentrations (1, 2, 5 and 8% (w/v)).

observable fact was that the profiles evolved with an increase in feed peptide concentration. The peptide population with retention times between 24 and 30 min (corresponding to hydrophobic peptides) increased. Indeed, these peptides seemed more abundant as the feed peptide concentration increased. According to the mass spectrometry analyses, 14 peptides were recovered at concentrations of 2, 5 and 8% (w/v) (Table 1). By contrast, eight other peptides were not present in the recovery compartment at a feed concentration of 1% (w/v). According to a previous study (Koumfieg Noudou et al., 2016), the selective transport of peptides rich in arginine and/or lysine was improved with increase in feed peptide concentration (Table 1). Despite their concentration increases, the same peptides were present in the recovery compartment whatever the initial feed concentration from 2%

(w/v). Hence, the peptide selectivity is not affected but the increase of peptide concentration in the feed solution would suggest a modification of recovered peptide concentrations among themselves.

3.3. Impact of feed concentration on the α 137–141 recovery

The α 137–141 peptide concentration increased throughout the experiment duration except for the 10 last minutes in the recovery compartment (Fig. 2e). The α 137–141 recovery was raised by increasing the initial feed peptide concentration: its concentration increased from 3.47 ± 0.42 , 4.63 ± 0.68 , 7.04 ± 0.68 to $13.1 \pm 0.50 \text{ mg.l}^{-1}$, respectively at 1, 2, 5 and 8%. This corresponded to a 3.7-fold increase between 1 and 8% (w/v) concentrations of hydrolysate. Nevertheless,

Table 1

Identified peptide sequences in the recovery compartment.

Peptides in **bold** were the peptides identified in the cationic recovery compartment using the feed peptide concentration at 1% (w/v). The other peptides were identified under the other feed peptide concentration conditions (2, 5 and 8%, w/v) in addition to the peptides in bold.

Position	Sequence	Molecular weight (Da)	pI	Described antimicrobial activity
α137–141	TSKYR	653	10.50	(Catiau et al., 2011)
α136–141	LTSKYR	766	10.47	/
α99–105	KLLSHSL	796	9.91	(Adje, Balti, Kouach, Guillochon, & Nedjar-Arroume, 2011; Adje, Balti, Kouach, Dhulster, et al., 2011)
α135–141	VLTSKYR	865	10.46	/
α99–106	KLLSHSLL	909	9.91	(Adje, Balti, Kouach, Guillochon, & Nedjar-Arroume, 2011; Adje, Balti, Kouach, Dhulster, et al., 2011)
β130–139	QKVVAGVANA	956	9.83	/
α134–141	TVLTSKYR	966	10.40	/
β32–40	VVYPWTQRF	1195	9.81	/
α34–43	LSFPTTKTYF	1204	9.74	/
α37–46	PTTKTYFPFH	1238	10.09	(Nedjar-Arroume et al., 2008)
β114–124	ARNFGKEFTPV	1247	10.18	(Adje, Balti, Kouach, Guillochon, & Nedjar-Arroume, 2011; Adje, Balti, Kouach, Dhulster, et al., 2011)
β31–40	LVVYPWTQRF	1308	9.83	/
α129–141	LANVSTVLTSKYR	1451	10.47	/
α34–46	LSFPTTKTYFPFH	1586	9.74	(Nedjar-Arroume et al., 2008)

the α137–141 recovery increase did not show the same tendency as the total peptide recovery, depending on the feed peptide concentration, as previously observed for cationic peptides in EDUF (Koumfig Noudou et al., 2016). Indeed, in the present study, the purities of the recovered α137–141 were 50.1 ± 1.61 , 29.2 ± 2.53 , 18.3 ± 1.17 and $18.5 \pm 0.74\%$ for feed hydrolysates of 1, 2, 5 and 8% (w/v). Consequently, the maximal enrichment factor in α137–141 was obtained at 1%, i.e. 75.9 ± 15.2 . It appeared from these results that, after a drop between 1 and 2%, the concentration factor reached a plateau at 5% and higher (w/v) with a final enrichment factor of about 24-fold: the increase in feed peptide concentration involved a higher α137–141 transport but not with the same increase factor.

Using a working pH of 9, most peptides contained in the hydrolysate were negatively charged. By contrast, the α137–141 peptide was positively charged and also has a small molecular weight (653 Da). These characteristic features facilitated its migration through the ultrafiltration membrane considering the highest α137–141 purities at low feed peptide concentrations. This result was in accordance with previous studies which demonstrated that the chitosan dimer was the most mobile in a mixture of several oligomers with larger sizes (Aider, Brunet, & Bazinet, 2008a, 2008b). In addition, more peptides can interact with the ultrafiltration membrane and migrate according to their charges when the feed peptide concentration is increased (Koumfig Noudou et al., 2016). This fact could explain the fact that the migration of other peptides decreased the α137–141 purities when using high feed peptide concentrations. Nevertheless, by increasing the feed peptide concentration, the α137–141 concentration available for migration in the feedstock became higher and consequently the α137–141 recovery concentration was the highest, since this peptide is the predominant one to migrate due to its physicochemical characteristics.

3.4. Relative energy consumption

The relative energy consumptions were assessed at the end of run time (Fig. 2f). The maximal energy consumption was observed for the 1% bovine hemoglobin concentration (w/v) with a value of $51.9 \pm 6.09 \text{ Wh.g}^{-1}$. The experiments with the 2% bovine hemoglobin concentration (w/v) allowed division of the relative energy consumption by more than two ($19.6 \pm 1.14 \text{ Wh.g}^{-1}$). This fact could be explained by the total peptide recovery which increased between these two feed conditions. Moreover, with a concentration equal to or > 5% (w/v) in bovine hemoglobin in the feed, the relative energy consumption reached a plateau with respective values of 4.68 ± 0.24 and $3.35 \pm 0.67 \text{ Wh.g}^{-1}$ for 5 and 8% (w/v). This decreasing trend was already observed previously for high concentrations in the feed compartment (Koumfig Noudou et al., 2016). However, in this previous

study, the plateau was reached at a lower concentration (2%) than in our case (5%). This difference could be due to the nature of peptides present in the hydrolysate, particularly their molecular weights and the number of sequences able to migrate through the filtration membrane. Indeed, the more peptides able to migrate at a low concentration of feed, the lower the relative energy consumption.

3.5. Antimicrobial activity of peptide fraction from EDUF

The recovered peptide fractions obtained for an 8% feed peptide concentration and containing the highest concentration of α137–141 were applied as food preservative at a peptides/meat ratio of 0.5% (w/w). The count results were expressed in survival rates to assess the influence of peptide fractions on the microbial growths.

In the meat, the total microbial population (Fig. 4a) evolved more slowly with the presence of additives for all the 14 days of experiment ($p \leq 0.05$). However, no significant difference was observed between the microbial growth in the meat treated by the standard α137–141 peptide and the EDUF fraction, whatever the day of storage. In general, the survival rates between the treated groups and the untreated group had a difference higher than $2 \log_{10}(N_0/N_t)$.

For the yeasts and molds (Fig. 4b), coliform bacteria (Fig. 4c) and lactic acid bacteria (Fig. 4d), the same evolutions were observed. Indeed, no-difference was shown between the two treated groups but high effects of additives were observed on all counted microbial populations with an effect always significant at day 14 of the experiment.

From these results, we can conclude that the recovered peptide EDUF fraction, using 8% feed peptide concentration, has an excellent potential to delay the evolution of bacteria present in the meat and, in general terms, to protect against microbial proliferation. Nevertheless, the observed no-different antimicrobial effect between the standard α137–141 peptide and the EDUF fractions provokes further thoughts. First, it was demonstrated that the EDUF is an efficient method to select a fraction enriched in α137–141 peptide with a maintained activity, as previously studied for some activity from various protein sources (Doyen, Beaulieu, Saucier, Pouliot, & Bazinet, 2011; Firdaous et al., 2010; He et al., 2016). Second, due to the α137–141 purity in the tested sample ($18.5 \pm 0.74\%$) and by comparing with the second test (pure α137–141), it appears that other peptides took part in this antibacterial activity. From fourteen identified peptides of this tested fraction (Table 1), six peptides were previously identified as antimicrobial peptides: α137–141 (Catiau et al., 2011), α99–105 and α99–106 (Adje, Balti, Kouach, Guillochon, & Nedjar-Arroume, 2011), α37–46 and α34–46 (Nedjar-Arroume et al., 2008) and β114–124 (Adje, Balti, Kouach, Guillochon, et al., 2011). Moreover, the peptides called α136–141, α135–141, α134–141 and α129–141 are potential

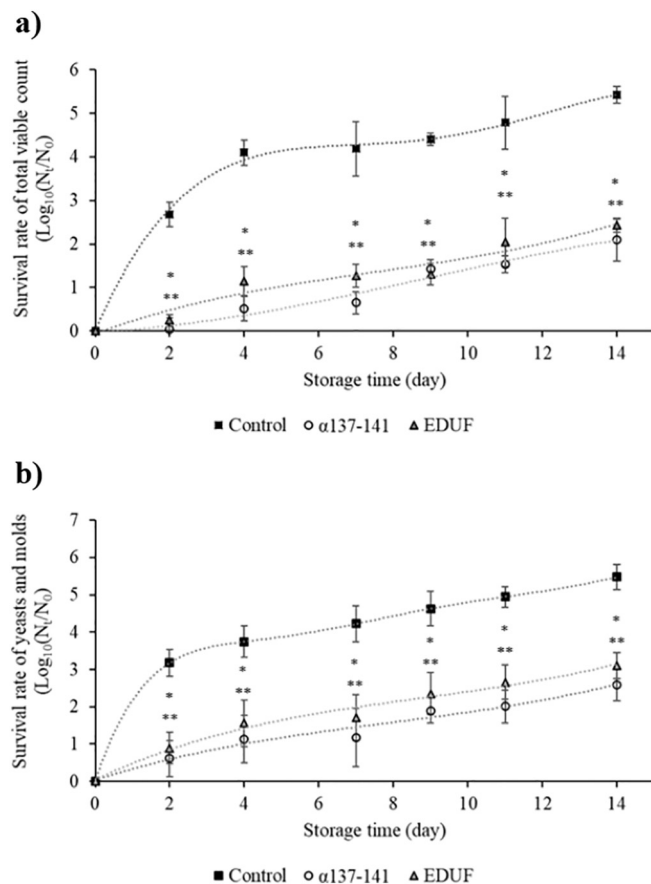


Fig. 4. Survival rates of total viable bacteria (a), coliform (b), lactic acid bacteria (c) and yeasts and molds (d) contained in meat treated with peptide fraction from EDUF, the standard α 137–141 peptide and without treatment. p-value > 0.05 (noted by *) indicates a significant difference between the EDUF condition and the control. A p-value > 0.05 noted by ** indicates a significant difference between the control and the α 137–141 standard peptide condition.

precursors of α 137–141. All of them were derived from the main precursor (α 107–141) containing the minimal antimicrobial determinant (KYR) and having a described antibacterial activity (Catiau et al., 2011). Hence, it is supposed that the peptides α 136–141, α 135–141, α 134–141 and α 129–141 would be antimicrobial peptides.

However, the α 137–141 peptide should make one of the highest contributions to the EDUF fraction activity based on the minimum

inhibitory concentration of the other peptides found in the literature. Briefly, the α 137–141 minimal inhibitory concentrations (MIC) were between 1 and 9 μ M against four tested strains: *Micrococcus luteus*, *Listeria innocua*, *Escherichia coli* and *Salmonella enteritidis* (Catiau et al., 2011). The α 137–141 peptide was the most efficient against *Listeria innocua*, *Escherichia coli* and *Salmonella enteritidis* with MIC values relatively close and of the same magnitude as the α 37–46, α 34–46 and β 114–124 peptides (between 1 and 12 μ M) (Adje, Balti, Kouach, Dhulster, et al., 2011; Nedjar-Arroume et al., 2008). For the α 99–106 and the α 99–105, and whatever the tested strains, the recorded MIC values were between 34 and 78 μ M, indicating lower antimicrobial activities compared to the α 137–141 which had MIC between 1 and 9 μ M for all tested strains (Adje, Balti, Kouach, Dhulster, et al., 2011). In addition, based on a previous study (Przybylski et al., 2016) where pure α 137–141 was compared to the synthetic butylated hydroxytoluene (BHT) in terms of antimicrobial activity, the fact that the α 137–141 and the EDUF peptide fraction had similar antimicrobial activities suggests that the EDUF peptide fraction would have an antimicrobial effect close to that of synthetic butylated hydroxytoluene (BHT).

3.6. Antioxidant activity of peptide fraction from EDUF

Fig. 5 shows the evolution of lipid oxidation in the meat under three conditions (without treatment, with the recovered EDUF peptide fractions using 8% feed peptide concentration and with standard α 137–141). For all meat samples, the TBARS values increased throughout the fourteen days of experiment. For all days, the control (i.e. without additive) had higher values than all tested conditions, particularly at the end of storage (day 14) with a p-value $\leq$ 0.001 for the two treated groups. During the first nine days of experiment, no significant difference between the meat treated with EDUF fraction and the meat treated with α 137–141 standard was observed (p-values $\geq$ 0.09). However, a significant difference between these two groups was observed at days 11 and 14 with respective p-values of 0.04 and 0.01. Indeed, the TBARS values were lowest for the group treated with α 137–141. At day 11, the TBARS value for the group treated with EDUF peptide fractions was 0.766 ± 0.079 mg MDA/kg meat and it was 0.597 ± 0.029 mg MDA/kg meat for the group with standard α 137–141. At day 14, the TBARS values were, respectively, of 0.871 ± 0.080 and 0.620 ± 0.053 for the groups with EDUF peptide fraction and with only the standard α 137–141 peptide. In comparison, the control group without additive had TBARS values reaching 1.54 ± 0.068 and 1.75 ± 0.122 mg MDA/kg meat at the end of storage (day 14). Consequently, the reduction of lipid oxidation in the meat was 50% for the EDUF fraction and about 60% for the standard α 137–141 peptide. This difference could be explained by the presence

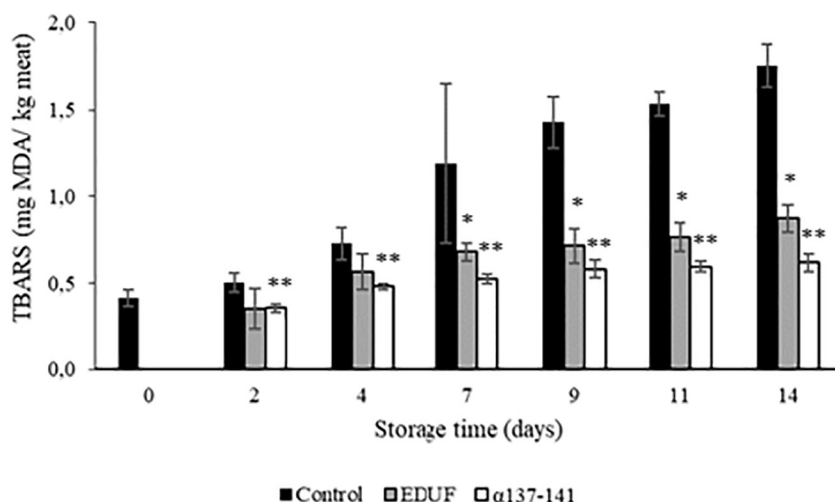


Fig. 5. Comparison of lipid oxidation of the meat treated with peptide fractions from EDUF, α 137–141 standard peptide against meat without additive during storage under refrigeration.

p-value > 0.05 (noted by *) indicates a significant difference between the EDUF condition and the control. A p-value > 0.05 noted by ** indicates a significant difference between the control and the α 137–141 standard peptide condition.

of other peptides in the fractions which reduced its antioxidant activity in comparison with α 137–141 peptide alone. Also, based on previous published data, the observed antioxidant effect of the EDUF peptide fraction was close to that of BHT (Przybylski et al., 2016).

4. Conclusion

In this study, EDUF was a suitable and sustainable technology to enrich a fraction in the α 137–141 antimicrobial peptide from the crur hydrolysate, with an obtained purity of 18% corresponding to an enrichment factor of 24 compared to the initial hydrolysate at a feed peptide concentration of 8%. Moreover, this condition involves minimal energy consumption that stimulates the peptide separation with high peptide concentration in feedstock. Finally, the excellent microbial growth inhibition and the antioxidant capacities of the EDUF peptide fraction showed that it could be a promising potential food preservative and substitute for the synthetic additives widely used to protect meat during its storage or distribution.

This new application of EDUF and the production of antimicrobial/antioxidant peptide or peptide fraction, in an energy efficient approach, from waste, fits perfectly with the concept of circular economy. Indeed, hemoglobin from blood, once hydrolyzed, allows the bioproduction of antimicrobial and antioxidant peptides that can then be used in meat or meat products to increase their conservation and safety.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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